

How much does the map of science depend on the embedding model?

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Abstract

A science map is useful only if its conclusions can be interpreted with some confidence in how the map was made. Here, ten embedding models represent the same fixed corpus of 500,000 OpenAlex records from 26 fields and 2000–2024. Matched panels compare field centres, internal structure and nearest neighbours, then vary text input and representation settings. Broad structure is shared, but models retain only 30.3% of the 25 nearest neighbours on average within field-period groups. Removing abstracts changes neighbours about as much as changing the model; removing titles changes less. An exploratory comparison of all 325 field pairs finds persistent opposite orderings across models in 262 pairs for angular spread and 221 for effective linear dimension. Requiring relative differences above 5% leaves 96 and 177. Alternative measures preserve many oppositions, while centring and pooling affect their persistence. These are geometric findings, without an external test of thematic correctness. A comparison can survive repeated article selection and still reverse when the model changes. Robustness assessment should follow the particular conclusion drawn from a science map, alongside its overall agreement with other maps.

Keywords: science mapping; document embeddings; robustness; neighbourhood agreement; geometric comparisons.

1 Introduction

A science map helps a reader decide which papers to place together and how to compare research areas. With text embeddings, that organisation starts when a model turns each paper into a vector. Distances between vectors then provide the relationships on which the map rests. This approach has been used to map biomedical literature and investigate the disciplinary structure of physics (González-Márquez et al., 2024; Constantino et al., 2025). Its usefulness leaves a practical question: if another model represented the same papers, which conclusions would still hold?

Science mapping has faced this problem before. Gläser et al. (2017) asked how far a reconstructed thematic structure reflects scientific knowledge or the method used to recover it.

Boyack and Klavans (2020) compared text, citation and hybrid relations and found substantial changes in cluster membership despite similar aggregate accuracy scores. A close overall score can therefore conceal a difference that matters to a reader: the methods may group different papers together.

This comparison keeps the articles fixed and follows the consequences of changing their representation. The corpus contains 500,000 publications labelled English in OpenAlex, published in 2000–2024 and assigned to 26 broad Fields. Ten embedding models represent those papers. The analysis moves from relations between field centres to the structure within fields and the neighbours of individual articles. Matched subsets vary the input text and the rule used to reduce model output to one vector per paper. A later geometric extension tests whether a comparison between two fields keeps its direction when the model and article selection change.

The models share broad relationships, yet differ in local neighbours and in comparisons between fields. A geometric ordering can recur across article selections in one model while the opposite ordering recurs in another. Its magnitude and sensitivity to processing determine how much to make of that disagreement. This places a limit on what repeated selection alone can establish. Agreement between representations identifies shared structure; assessing whether that structure captures the subject matter requires further evidence.

2 Background and research questions

2.1 Representation choices in science mapping

Several studies already show why a document representation needs an explicit purpose. Constantino et al. (2025) assessed text and graph representations against a hierarchical classification of physics papers, including representations of titles and abstracts. In biomedicine, González-Márquez et al. (2024) compared models and output-reduction rules before choosing a representation for a large literature map. Their criterion was how often nearby papers shared field labels. In both cases, the interpretation of performance depends on the criterion used to judge it.

Naming the model leaves several choices unspecified. A title and an abstract supply different information; models also accept different amounts of text. Some require a further decision about how to combine token representations into a document vector. This operation, called pooling, can average the token outputs or use one designated position. Singh and Singh (2022) showed that scientific language models can change their retrieval behaviour under short queries and changes in wording. A comparison of science maps should make these input and representation choices visible.

Geometry and retrieved neighbours have also been compared together. Caspari et al. (2024) used both across retrieval datasets, while Imel and Hafen (2025) examined local density and asymmetry across document embeddings and disciplines in relation to citation rates. The question pursued here is how far broad, local and geometric conclusions persist across a fixed model panel. Matching articles, text conditions, representation settings and candidate papers makes it possible to specify which change each comparison addresses.

2.2 Agreement, stability and interpretation

Agreement describes what two models share when they represent the same articles. Stability asks whether a stated result survives a stated change, such as selecting other articles from the corpus. These can come apart: a field ordering may be stable within each model and differ between models. Thematic validity asks a further question about whether the represented relationships describe the research appropriately. In invasion biology, Held and Velden (2022) showed why interpreting algorithmic groups requires attention to the different relationships a mapping method combines. Repetition and consensus alone cannot settle that interpretation.

Here, *shape* refers to relationships in the original embedding spaces, before any projection into two dimensions. Field centres describe broad organisation. Comparisons within fields retain article-level structure, and neighbour agreement identifies whether the same papers stay closest to a given article. Angular spread and effective linear dimension describe further geometric properties: how widely the vectors open out and how their variation occupies different directions. Their names need to stay close to those operations. Greater spread does not establish greater topic diversity, and effective linear dimension does not count topics. A change in one property also need not imply a change in every other part of the map.

2.3 Research questions and scope

Three questions organise the analysis:

- RQ1.** Which structural relationships are shared across embedding models, and how does agreement vary between field centres, articles within fields and local neighbours?
- RQ2.** How much do text input and pooling choices change geometry and neighbours relative to changing the embedding model?
- RQ3.** Which geometric comparisons between fields persist, reverse or remain unresolved across models and article selections, and how large are the differences?

The questions describe the selected corpus and model panel. The third arose after the initial comparisons, in an exploratory extension whose sequence is recorded in the methods and

supplement. OpenAlex labels organise the analyses, with their known errors retained. They provide no independent test of thematic correctness. The objective is to establish which conclusions persist within this design; recovering or ranking models against a single correct map would require a different form of evidence.

3 Data and comparative design

3.1 Corpus and analytical panels

The corpus contains 500,000 distinct OpenAlex work IDs, collected and cleaned by 15 September 2026. OpenAlex provides the metadata and texts (Priem et al., 2022); its core catalogue was restricted to articles, reviews and conference papers published in 2000–2024, labelled English, with a non-empty title, a reconstructible abstract of at least 50 words and a known primary Field. Retraction and paratext flags were excluded. Clear language and content failures were removed, while uncertain cases and possible duplicate versions were retained with flags. Each record’s primary-topic classification was frozen at extraction. These automatically assigned labels use text, citation and journal information (OpenAlex, 2026); they organise the analysis rather than supply an independent standard of correctness.

A general sample of 400,000 eligible records was drawn through seeded random API requests. An additional 100,000 records reinforced the smallest Field-period groups by repeatedly allocating places to the least populated group. Large groups were not reduced to the size of the smallest. The resulting 130 groups cover 26 Fields and five periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Every group contains at least 2,165 records. Global corpus comparisons use the general sample; group comparisons can use both cohorts. Equal-weight averages across groups describe a balanced comparison, not the proportions of world research.

The panels in Table 1 keep distinct comparisons apart. For the three text inputs, 52,000 articles are divided equally across Field-period groups, including the earlier pilot of 26,000. A different panel of 52,000 articles compares each model’s native input with the same literal fragment, chosen to fit every tokenizer. Equal-size group comparisons and matched neighbour searches address further sources of variation. Article identity is held constant between models in every comparison. Supplement S1 gives the extraction rules, cleaning steps, counts and retained flags.

Table 1: Corpus and analytical panels.

Panel	Articles and representation	Comparison and scope
Original corpus	500,000 IDs: 400,000 base + 100,000 supplement. Ten models; title + abstract.	26 Fields, five periods. Global summaries use the base; the full corpus is not population-proportional.
Three text inputs	52,000 IDs; 400 per Field-period, 2,000 per Field. Title, abstract and both.	Same articles and candidates between models. Includes the 26,000-article pilot.
Common fragment	A different 52,000-ID panel; 400 per Field-period.	Paired native/shared content; model-specific tokenization. Distinct from the three-input panel.
Matched group size	256 articles per centre; 26 Fields, 217 Subfields, or 26 Subfield centres.	Within groups: 26 Fields and the same 183 Subfields at 128, 256 and 512 articles. Different objects from centres.
Paired searches	217 Subfields; 50 fixed queries each (10,850 total). Ten candidate selections.	256 candidates in each scope, including all 50 queries. The Field pool is conditioned on these queries.
Geometry / Field pairs	2,000 per Field; twenty halves of 1,000 and five selections of 2,000.	325 Field pairs and two properties; 26 saved conditions. Alternative metrics have different coverage.

Rows overlap and must not be added. Periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Each model comparison uses the same IDs. Query articles exclude themselves during neighbour search. The 1,300 technical test articles are not an additional scientific panel. Full filters and quotas: Table S1.

3.2 Models and text representations

Ten models cover scientific-document, general sentence and word-level biomedical/scientific representations (Table 2). This is a fixed panel chosen for its relevance to scholarly representation and contrasting training approaches, rather than a random sample of encoders or a ranking of the latest systems. Checkpoints, tokenizer revisions and text formats were frozen before inference. Each model encoded the same title and abstract in its customary format and with its native input limit. MiniLM’s 256-token limit was retained as primary; a separate 512-token variant tests that choice. Native truncation affected 49.58% of MiniLM inputs and 20.16% of MPNet inputs.

For BERT, SciBERT, BioBERT and PubMedBERT, the principal document vector is the mean of last-layer, non-padding token states, including special tokens. These models were not trained specifically for document similarity. Their first-position state (CLS, before the pooler) and final non-padding state (SEP) provide alternative readouts. The mean rule was specified before the primary comparisons and retained after the early sensitivities; it was not selected to maximise agreement. The other six models use their documented representation recipes. Title-only and abstract-only inputs use the same 52,000 IDs as the combined-input comparison. Native limits mean that this tests practical ways of using the models, without isolating architecture from text length.

Table 2: Embedding models and primary representation settings.

Model	Dim.	Tokens	Readout	Reference
SPECTER	768	512	CLS	(Cohan et al., 2020)
SPECTER2	768	512	CLS + adapter	(Singh et al., 2023)
SciNCL	768	512	CLS	(Ostendorff et al., 2022)
SciBERT	768	512	Mean	(Beltagy et al., 2019)
BERT	768	512	Mean	(Devlin et al., 2019)
MPNet	768	384	Mean	(Reimers and Gurevych, 2019)
MiniLM	384	256	Mean	(Reimers and Gurevych, 2019)
PubMedBERT / BiomedBERT	768	512	Mean	(Gu et al., 2022)
BioBERT	768	512	Mean	(Lee et al., 2020)
SimCSE	768	512	CLS, pre-pooler	(Gao et al., 2021)

MPNet and MiniLM use all-mpnet-base-v2 and all-MiniLM-L6-v2, respectively. Limits count tokens including special symbols. Mean: last-layer average over non-padding positions, including special symbols. CLS: first-position state before the pooler; SPECTER2 adds its proximity adapter. For MPNet/MiniLM, the reference describes the Sentence-BERT framework, not the exact check-point training. Readout variants are not additional models. Full revisions, formats and truncation: Table S2.

3.3 Objects and measures

All measurements use the original embedding dimensions, without a two-dimensional projection. Article vectors are first scaled to unit length. Shape agreement is measured primarily by corrected linear centred kernel alignment (CKA), which compares relationships among the same identified articles (Kornblith et al., 2019; Murphy et al., 2024). Higher values indicate greater agreement. The finite-sample correction reduces a known source of inflated alignment; it does not make the complete ratio an exactly unbiased estimator. Procrustes

alignment and correlations between distance ranks provide checks with different sensitivities. Their definitions and the correction formula appear in Supplement S2. These related measures are not independent votes about validity (Williams et al., 2021; Williams, 2024).

A group centre averages its unit-length article vectors. Comparing centres therefore describes the organisation of groups, whereas comparing their constituent articles preserves variation within them. The random-group control asks how much centre agreement can arise from averaging without the observed labels. It keeps group sizes and publication periods fixed and uses the same random allocation in every model.

Neighbour agreement is the fraction of the k closest articles shared by two models. The main choice is $k = 25$, with 10 and 50 as checks. Searches use cosine similarity and exclude the query article itself. The candidate pool is stated for each result: all articles in a Field-period group, an equal-size subset, a matched Field/Subfield pool, or the general corpus. An exact blockwise calculation avoids storing every pairwise distance. Independent full-sort checks verify saved neighbour lists. Neither overlap nor its simple chance-adjusted form removes all effects of candidate composition or size.

Two additional properties describe each Field geometrically. *Angular spread* is the median angle between distinct article vectors. *Effective linear dimension* is the participation ratio of the centred covariance matrix C , $PR = (\text{tr } C)^2 / \text{tr}(C^2)$. It is larger when variation is distributed across more directions. Angular quantiles, angles to the centre, entropy-based effective rank and the number of directions containing 80% of variance serve as alternatives. The latter two use the same covariance spectrum as PR; they do not provide independent thematic validation. Graph-connection measures were also tested, but their synthetic counterexamples prevented an interpretation as general fragmentation (Supplement S6).

3.4 Matched comparisons and stability

The strict Field/Subfield neighbour comparison fixes 50 query articles in each of 217 Subfields. Both searches contain 256 candidates, including those same 50 queries, and identical period counts. Ten selections vary the remaining 206 candidates within the Subfield or its parent Field. Thus, the broader search is conditioned on the queries; it is not an unrestricted draw of 256 Field articles. The design covers 10,850 queries and holds search opportunity constant while varying its scope.

For geometric Field comparisons, every unordered pair of the 26 Fields is evaluated: 325 pairs for each property. A model gives a persistent direction only if that direction holds across the primary 2,000 articles per Field, twenty half samples of 1,000 and five additional selections of 2,000 from the same corpus. All-ten agreement requires the same persistent direction in every model. Opposition requires at least two fixed models with persistent opposite directions. Remaining pairs are unresolved. This does not mean equality. Relative

magnitude is $2(B - A)/(A + B)$, where A and B are the two Field values; minima of 0, 1, 5 and 10% test sensitivity to small differences.

The repeated selections test sensitivity inside the frozen corpus. Because the subsets overlap, their ranges cannot be read as confidence intervals from independent population samples. The operational screens flag excessive median shifts or selection ranges; the supplement keeps each rule and denominator separate. Model pairs share models, and Field pairs share Fields. Their counts are consequently descriptive, and no significance test treats them as independent observations.

3.5 Sequence of checks

The work developed through successive comparisons. Shape and neighbours came first, followed by the expanded input panel, matched Subfield searches and case atlas. The morphology extension was exploratory; its all-pairs summary followed inspection of the initial examples. Rules were recorded locally before each corresponding new calculation, without externally preregistering the entire study. Random-group centres and the fixed-neighbour-fraction check were among controls added after earlier results. Supplement S2 records this sequence. Supplements S3–S7 retain the checks on time, discipline composition, model traits, source errors and processing, including those that restrict the interpretation.

4 Results

4.1 Shared broad structure and within-field variation

Models agree strongly about the arrangement of group centres. With 256 articles contributing to each centre, mean corrected CKA is 0.939 for the 26 Fields and 0.911 for 217 Subfields (Figure 1A). Restricting the latter to 26 Subfield centres gives 0.902. Random groups with matched sizes and dates also show positive agreement, around 0.63–0.64, but do not recover the observed level. Averaging therefore contributes to the comparison, yet is insufficient to account for the agreement associated with these labelled groups.

Inside the groups, agreement has a different level. Mean CKA is around 0.63 for Fields and 0.64 for the same 183 Subfields eligible at each size (Figure 1B). Increasing the article count from 128 to 512 changes these averages little. Finer labels therefore do not imply lower agreement in this comparison. The centre and within-group panels describe different objects, so the gap between them cannot isolate a causal effect of scale.

The alternative measures broadly support the ordering of model-pair agreements. Across the original 130 Field-period groups, the mean rank correlation with corrected CKA is 0.891 for Procrustes and 0.881 for distance ranks. Their agreement is incomplete, and the original

selection screen retains 50 alerts among 5,850 model-pair/group comparisons (Table S4). These qualifications concern particular comparisons; an overall mean does not erase them.

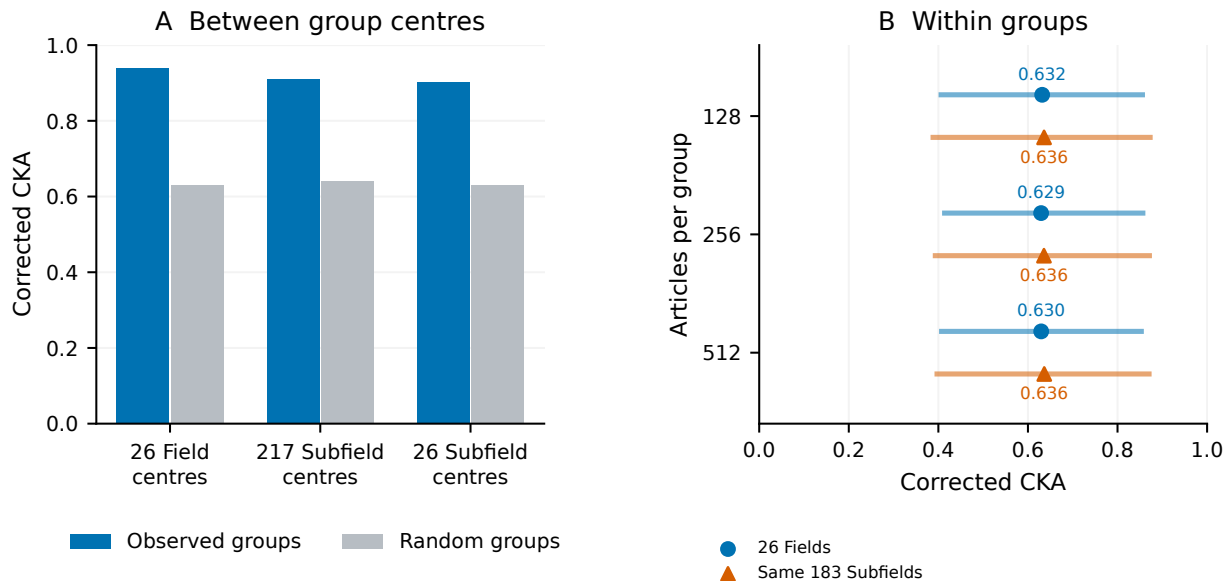


Figure 1: Agreement between group centres and within groups. (A) Mean corrected CKA across the fixed model pairs and saved centre comparisons; each centre contains 256 articles. Random groups preserve sizes and dates. The 26-Subfield condition controls the number of centres. (B) Within-group agreement at three article counts, for 26 Fields and the same 183 eligible Subfields. Points and labels show means; lines span the 2.5th–97.5th percentiles across group–model-pair comparisons (1,170 for Fields; 8,235 for Subfields). These are descriptive spreads, not confidence intervals; model pairs are not independent. All CKA axes span 0–1. The panels compare different objects, so their difference is not an estimated causal effect.

4.2 Neighbour agreement under matched comparisons

Across the full Field-period searches, two models retain an average of 30.3% of the 25 nearest neighbours: about 7.6 articles out of 25. This average gives each group and model pair equal weight. Restricting every group to 2,048 common candidates raises overlap to 31.9%. In the separate search across the 400,000-article general corpus, 13,000 balanced queries retain 17.5%. The last result changes both the scope and the number of available candidates, so it cannot by itself establish weaker agreement across disciplinary boundaries.

The matched Field/Subfield experiment addresses that ambiguity directly (Figure 2). With the same queries, dates and 256-candidate size, Subfield agreement is lower in 127 of 217 cases and higher in 90. The mean difference is small, at -1.35 percentage points. For 50 neighbours, the split becomes 109 lower and 108 higher. A narrower search can preserve more

neighbours or fewer, depending on the group and the chosen neighbourhood size. There is no universal loss of agreement from Field to Subfield.

Shape and neighbour agreement describe related changes. Their descriptive rank correlation across the original 45 model-pair means is 0.920. There is still a practical reason to report both: CKA summarises distributed relationships, while a neighbour list identifies the articles a reader would actually encounter. The two measures expose different consequences of representation choice without being unrelated dimensions of reliability.

The atlas makes those local differences inspectable. Its Subfield examples are selected by a recorded persistence rule across 27 combinations of size, readout and neighbour count. Article-level examples then show links that persist across models alongside links whose inclusion varies sharply (Supplement S4). Some extreme cases expose source problems: a history book appears under Physics, an infection-related record under Political Science, and an announcement remains in the corpus. Comparison with the original records confirms that these are source or classification issues, rather than mismatched embedding rows. The examples are exploratory and deliberately selected; they estimate neither the prevalence of such errors nor the accuracy of the links.

4.3 Text input and representation choices

Removing the abstract can change a representation about as much as changing the encoder. On the same 52,000 articles, the mean shape change, measured as one minus corrected CKA, is 0.370 when changing model and 0.366 when retaining the model but using only the title. The corresponding fractions of neighbours replaced are 68.8% and 70.7% (Figure 3). These close averages do not establish equivalence. They show that a seemingly simple input choice can be comparable in scale to the model choice in this design.

Removing only the title changes less. Abstract-only inputs have a mean shape change of 0.027, yet replace 22.1% of the 25 neighbours. A small change in overall geometry can therefore still alter the papers returned locally. The model-specific results also vary substantially: title-only neighbour replacement ranges from 51.7% for MPNet to 84.2% for SciBERT (Table S5). A single pooled mean would conceal those differences.

The pooling rule can alter the comparison used to assess input changes. With mean pooling as the principal readout, removing the title has a relatively modest effect, and removing the abstract slightly exceeds model change in average neighbour replacement. Using CLS or SEP for the four word-level BERTs reverses that latter ordering. Four representations change together, including the reference for the cross-model comparison. This prevents an isolated causal reading of pooling and gives no basis for choosing the readout with the highest agreement.

The larger input panel reduces sensitivity to article selection. Using 100 comparable selections at each size, the shape-contrast screen drops from 72 alerts among 520 model–Field–

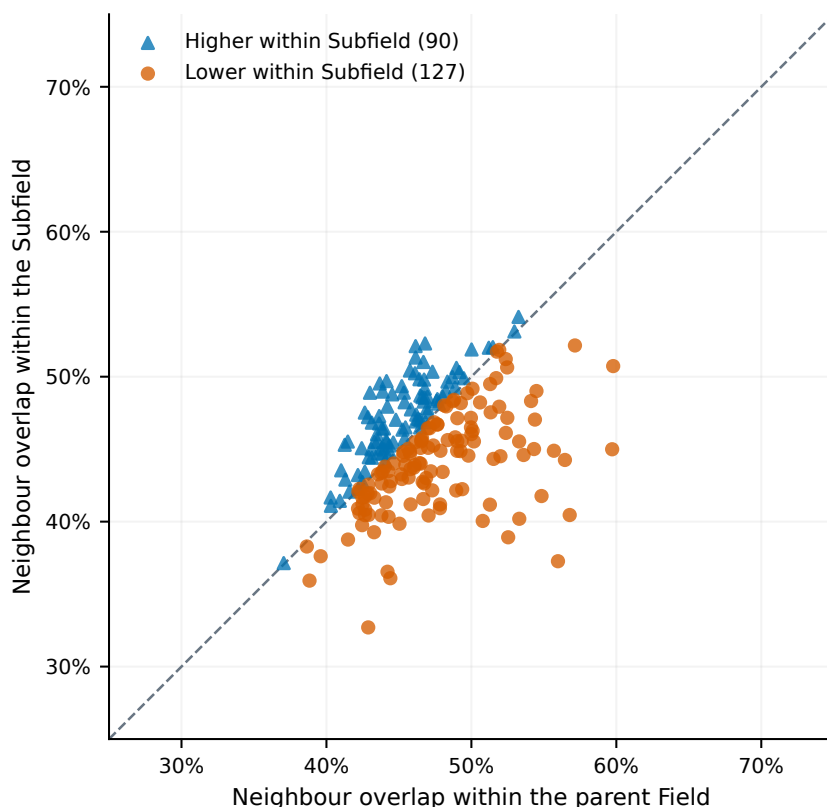


Figure 2: Neighbour agreement in matched Field and Subfield searches. Each point is one of 217 Subfields, averaged over model pairs and ten candidate selections. Both scopes contain the same 50 fixed queries within 256 candidates and matched dates; self is excluded. The broader Field pool is conditioned on these queries. Points above the diagonal have higher overlap in the narrower search; those below have lower overlap. Dispersion describes group differences, not confidence intervals.

input contrasts at 26,000 articles to five at 52,000. The earlier 20-selection check gives 31 and four, respectively (Figure S1). Both records are retained. Passing this screen does not mean that a contrast differs from zero, nor does it certify all neighbour results or readout variants. The common-fragment and MiniLM-length checks further distinguish unequal text exposure from a coding error: the original MiniLM limit was its documented setting, not an implementation fault.

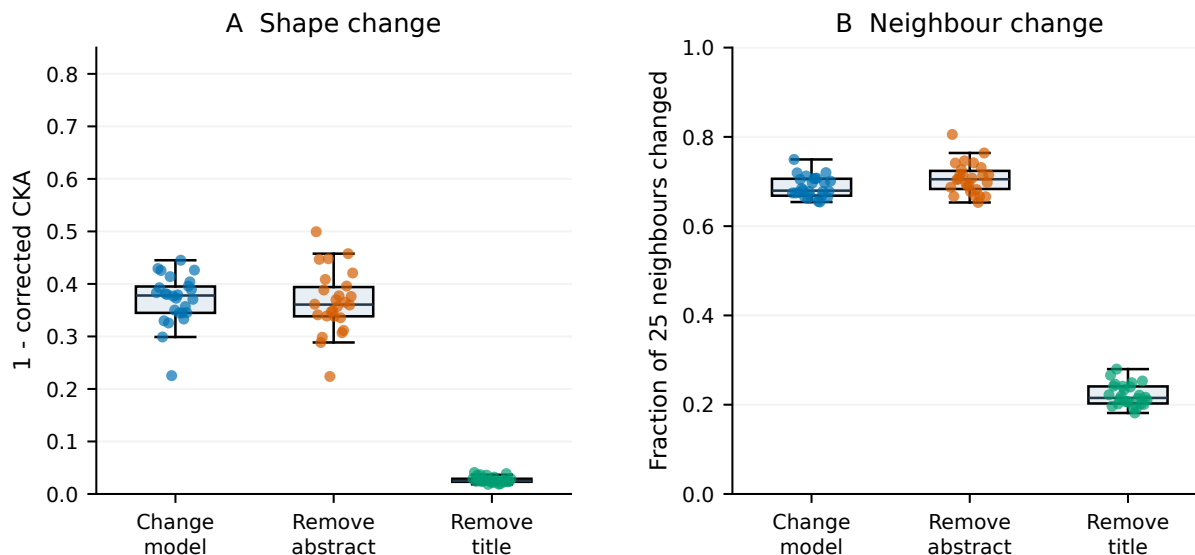


Figure 3: Changes associated with model and text input. The same 52,000 IDs provide 2,000 candidates per Field. The reference input is title plus abstract: “remove abstract” leaves the title, and “remove title” leaves the abstract. (A) One minus corrected CKA. (B) Fraction of the 25 neighbours replaced. Each point is a Field mean over the fixed model panel. Boxes show medians and interquartile ranges, with 1.5-IQR whiskers; all Field points are displayed. These are not confidence intervals. Native text limits are retained.

4.4 Persistence and reversal of geometric field comparisons

The geometric extension follows an individual claim about the map: whether one Field has a higher measured property than another. The first check changes the articles while retaining the model. Angular spread has no alerts among the 260 model–Field cases in either the half-sample or equal-size screen. Effective linear dimension has four half-sample alerts and none across the five equal-size selections (Table S12). The latter result does not remove the four retained alerts.

Stability within a model does not ensure agreement between models. Among all 325 Field pairs, angular spread has a persistent common direction in all ten models for 42 pairs, persistent opposition for 262 and an unresolved outcome for 21. Effective linear dimension has corresponding counts of 54, 221 and 50 (Figure 4). Opposition means that at least two particular models repeatedly support opposite directions across all 26 saved selection conditions. It does not require every model to disagree, and unresolved pairs can contain partial agreement.

Many angular differences are small. Requiring each opposing direction to exceed a symmetric relative difference of 5% reduces angular opposition to 96 pairs; at 10%, 19 remain. Effective

linear dimension retains 177 and 126 oppositions at those cutoffs. Thus, the direction-only totals should not be read as counts of equally large disagreements. The minimum is an explicit sensitivity choice, not a universal definition of a meaningful scientific difference. Figure 4 retains all 325 pairs in each row so that the growing unresolved category remains visible.

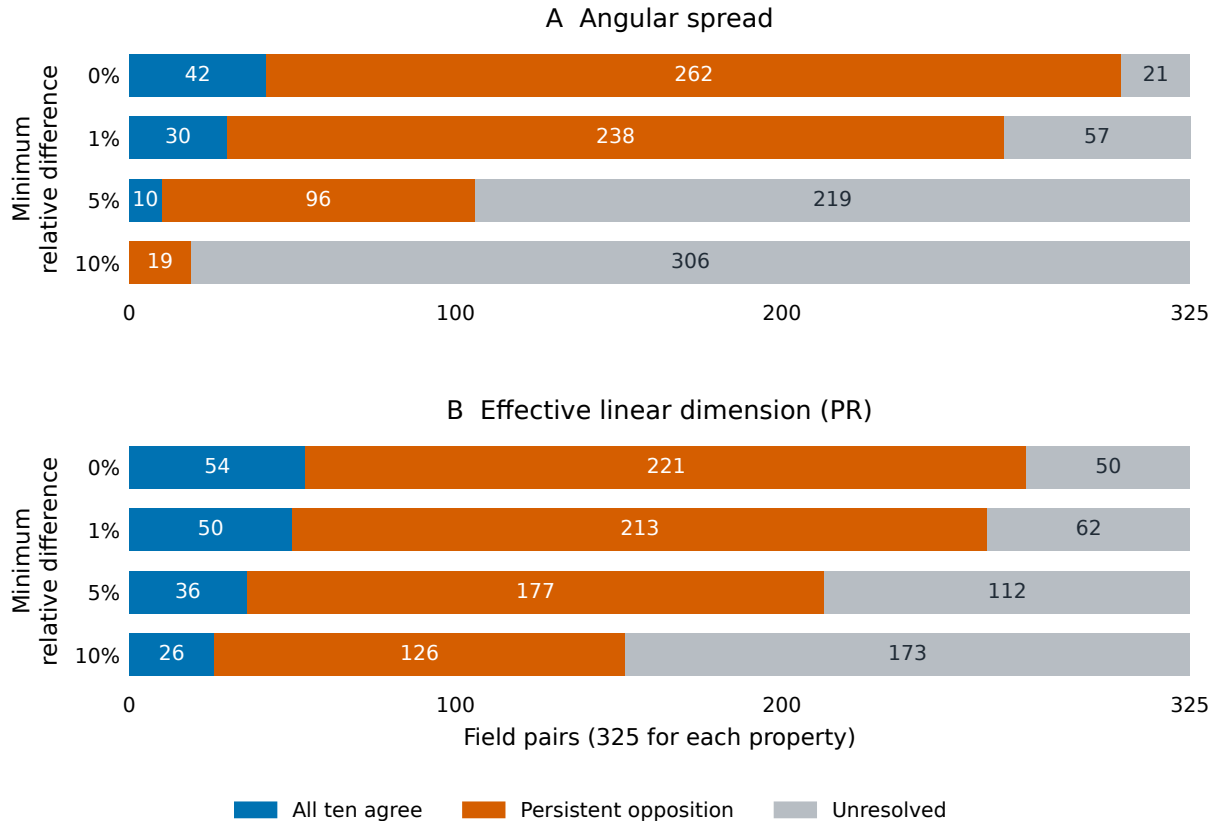


Figure 4: Persistence, opposition and magnitude of Field comparisons. Each row contains all 325 Field pairs for one property and ten models. A model direction must persist across the primary selection, twenty half samples and five additional equal-size selections from the same corpus. All-ten agreement requires a common persistent direction; opposition requires at least two fixed models with persistent opposite directions. All remaining pairs are unresolved. Magnitude uses $2(B - A)/(A + B)$. The four minima are operational cutoffs, not significance or accuracy thresholds. The 26 conditions overlap; point controls are reported separately.

Alternative measures preserve much of the pattern. Requiring the same opposing models to keep their directions across the angular alternatives retains 225 of the original 262 oppositions. The spectral alternatives retain 196 of 221. Their coverage differs: angular alternatives have all 26 selection conditions, whereas entropy-based rank and the 80%-variance count are available for three. These alternatives also share much of their mathematical construction. Their agreement supports a geometric description without turning it into a measure of topic diversity.

The result is not confined to the four word-level BERTs. Restricting the panel to SPECTER, SPECTER2, SciNCL, MPNet, MiniLM and SimCSE still yields 231 angular and 160 effective-dimension oppositions at zero minimum. A smaller panel has fewer opportunities to contain opposing models, so the change in counts does not isolate an effect of training objective. All model omissions and the complete pair classifications accompany Tables S14–S15.

Processing choices set a further boundary. After subtracting the global centre and renormalising, the same original opposing model pair and directions survive for 145 of 262 angular oppositions and 218 of 221 effective-dimension oppositions in the matched point check. Angular ordering is therefore more dependent on that choice. Pooling changes can also reverse comparisons (Table S16). These are point sensitivities with matched references, not another set of 26-repeat confirmations. The retained conclusions concern explicitly defined representations, rather than properties expected to survive any transformation.

The connection measures did not support the proposed interpretation as fragmentation. Under some rules, elongated clouds and isolated extreme points produced a similar signal, and changing the sample size or neighbour count substantially changed the result. The equal-size selection screen retained 51 connection alerts among 260 cases. Supplement S6 reports these measures together with the failed construct checks. They cannot support a claim that one Field contains more separate thematic groups.

5 Discussion

5.1 From map agreement to the persistence of a claim

The broad structure shared by the models is a useful starting point for interpretation. Centre agreement exceeds the random-group reference, so the common organisation cannot be reduced to that reference alone. The Field-pair results show where the inference needs to go further. One model can repeatedly place two Fields in one order, and another repeatedly place them in the opposite order, as articles change. The relevant robustness test therefore concerns the relationship, neighbour set or Field comparison that the analysis intends to interpret.

This question continues the comparative problem described by Gläser et al. (2017); Boyack and Klavans (2020). An external classification, as used by Constantino et al. (2025), tests how a representation serves a particular criterion. Joint comparisons of alignment and retrieval, as in Caspari et al. (2024), establish which relationships models share. The contribution here connects those levels to the persistence and magnitude of particular Field comparisons on matched articles. Earlier work already examined representation choice and supplied the agreement measures on which this comparison builds.

Angular spread and effective linear dimension remain useful when interpreted as the geometric operations they measure. Angles describe separation; PR describes the allocation of variance across linear directions. Reading either as interdisciplinarity or semantic diversity requires evidence about the articles’ content. The same care is needed when interpreting embedding isotropy (Rudman et al., 2022). The failed fragmentation checks make the issue concrete: a measure can reliably distinguish vector clouds without identifying the substantive property attached to its name.

5.2 Implications for comparative science mapping

A useful way to assess robustness is to start with the claim a map will support. For retrieval, that requires stating the candidate pool and neighbour count. For a geometric comparison between Fields, it requires the representation rule, article balance and minimum difference. The input experiment adds a straightforward reporting obligation. A model name alone leaves the reader unable to reconstruct what was represented; the title/abstract choice, token limit and pooling rule belong beside it.

The additional comparisons also place limits on explanations of disagreement. Medicine has lower shape agreement than Energy, but is not consistently lowest for neighbours. The Medicine–Energy difference remains after balancing Subfield composition and after excluding BioBERT and PubMedBERT (Supplement S5). Neither check supplies a unique cause. Model families also look different under different readouts. Architecture, training material and objectives vary together in this panel, preventing their separate causal attribution.

The time comparison shows why the opportunity to find a neighbour matters. Mean shape agreement rises slightly from the first period to the last. Mean neighbour agreement falls by 0.87 percentage points with all candidates, but rises by 0.91 points when each group has 2,048 candidates. The direction changes with the search design. Current model parameters represent publications from different dates; they do not reconstruct the knowledge available at those dates. These comparisons cannot establish a historical convergence of science itself.

5.3 Limits and next validation

The corpus bounds the interpretation. It contains selected OpenAlex records with usable abstracts and predominantly English text. The reinforcement sample, balanced panels and equal-weight averages support comparisons between Fields, without estimating the world-wide distribution of research. The atlas shows that classification and content errors remain. Cleaning removes specified failures; it cannot certify every label. Recorded selections and frozen checkpoints permit inspection of the comparison, while repeated selection within the corpus leaves its coverage bias in place.

The model panel imposes a similar boundary. Its members share architectures, training sources and objectives, and several were not trained for document similarity. Checks using six similarity-oriented models and alternative readouts address particular concerns, without making the panel representative of all encoders. Persistent opposition identifies dependence on the models tested. It leaves open which ordering, if either, better describes the subject matter.

The geometric extension and all-pairs summary also followed earlier results. Their recorded sequence and complete tables make the choices inspectable, while preserving their exploratory status. A useful next study would select one consequential comparison and test it against independently assessed article relationships or a question grounded in a discipline. That would supply evidence about thematic interpretation, which further agreement calculations alone cannot provide.

6 Conclusion

The models share a broad organisation of the literature while differing in article neighbours and the geometric ordering of Fields. Input choice can matter as much as model choice. A comparison can also survive repeated article selection and still reverse between encoders. These results favour checking the particular claim a science map is used to support, alongside overall similarity. Angular spread and effective linear dimension allow such checks when definitions, magnitudes and processing sensitivities remain explicit. They describe the representations produced here. Further validation is needed before those geometric differences can be interpreted as properties of scientific knowledge.

Declarations

Author contributions. Alejandro Treny Ortega conceived the project and directed the research and manuscript preparation as an independent researcher. Computational and editorial assistance is disclosed below.

Competing interests. The author declares no competing interests.

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Data and code availability. The project repository is <https://github.com/aleetreney/Shape-of-Science-Robustness>. This draft is accompanied by the tables used for its figures and the editable manuscript sources. The frozen corpus, vectors and complete local execution records have not been deposited in a persistent public archive. Public code alone does not provide full reproduction of the present analysis. The access inventory and

remaining release requirements are stated in Supplement S8; no data deposit or persistent identifier is claimed here.

Use of AI-assisted tools. OpenAI Codex assisted with literature searches, software development and analysis, visualisation, and manuscript drafting and editing. The author is responsible for the content.

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